ST5202 — Applied Regression Analysis
Lecture Notes 4: Multiple Comparisons

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Outline

§1. Introduction to multiple comparison

§2. General exploration

§3. Tukey's and Dunnett's procedure

§4. Pre-specified contrasts

§1.1. Principle of multiple comparison

- To investigate the effect of a factor, we need to find not only whether it has any effects but also what the effects are. For a factor with a levels, it is equivalent to find out, in the contrast space $\mathcal{R}^{a-1}$, which contrasts are significantly nonzero.
- There are two situations of study:
  - (a) No pre-hypothesized contrasts are specified, a general exploring is called to investigate all contrasts in order to discover the significant ones.
  - (b) There are certain scientifically important contrasts which are pre-specified for the investigation. We only need to find out whether these pre-specified contrasts are significant but do not care about other ones.
- In the above studies, multiple tests are considered simultaneously. The principle of multiple comparison is to control the family-wise type I error rate over all the tests under consideration at a given level in order to avoid any artifacts to be taken as significant.

§1.2. Cherry picking

- We go to a cherry grove to pluck cherries, we choose the best ones and ignore the others. It is a human nature. In scientific research, there is also a tendency for researchers to pick up the results which meet their expectation but to ignore the others. This behavior is said to be cherry-picking. Cherry picking can distort the truth.
- Suppose we are concerned with the mean of a distribution which is indeed $N(0,1)$ but is unknown to us. If we take only one random number X from this distribution, if $|X| \geq 1.96$, we claim that the distribution has a non-zero mean, the probability that we commit a mistake is 0.05.
- But if we take 100 of them, say, $X_i, i = 1, \ldots, 100$, and as long as there is one such that $|X_i| \geq 1.96$, we claim the mean is non-zero, then, almost surely, we will commit a mistake. The probability that at least one of them with absolute value exceeds 1.96 is $1 - (0.95)^{100} = 0.9941$.
- Roughly, there are about 5 of them which have absolute value bigger than 1.96. These are the best cherries in the cheery grove $N(0,1)$. We should not make judgment on the cherry grove based on its best cherries.

§1.3. Artifacts and family-wise type I error rate

- In multiple comparisons, we investigate many contrasts from a family. Even if all those contrasts are indeed zero, some of them could be claimed significant by tests controlling individual type I error rate. Those contrasts are said to be apparently significant or are called artifacts.
- In order to avoid claiming artifacts as significant contrasts, the family-wise type I error rate must be controlled.
- Denote the family of contrasts by $\mathcal{C}$. For any $j \in \mathcal{C}$, let T_j be the test statistic and the decision rule to reject the corresponding null hypothesis be $T_j \geq c_j$ for a critical value c_j. The family-wise type I error rate is
$$P(\cup_{j\in\mathcal{C}}\{T_j \geq c_j\}).$$
- If for each individual contrast the type I error rate is controlled at α, i.e., $P(T_j \geq c_j) = \alpha$, then obviously,
$$P(\cup_{j\in\mathcal{C}}\{T_j \geq c_j\}) > P(T_j \geq c_j) = \alpha,$$
the family-wise type I error rate is not controlled at level α.
- In multiple comparison, the c_j's must be chosen such that the family-wise type I error rate is controlled at given level α.

§2.1. General exploration: the problem

- The problem of general exploration is to find out if there are any contrasts which are statistically significant.
- The problem is formulated as follows. Let $\boldsymbol{\mu}$ be the vector of the group (or level, cell) means. Let $\boldsymbol{c}$ denote a contrast vector and $\mathcal{C}$ denote the set of all possible contrasts. The general exploration is to test

$$H_0 : \boldsymbol{c}^\top \boldsymbol{\mu} = 0 \quad \text{for all } \boldsymbol{c} \in \mathcal{C}.$$

- The hypothesis can be expressed in terms of the basis contrasts. Let $\boldsymbol{\theta}$ be the vector of basis contrasts. The above hypothesis is equivalent to

$$H_0 : \boldsymbol{a}^\top \boldsymbol{\theta} = 0 \quad \text{for any} \quad \boldsymbol{a}.$$

- In the general exploring of main effects,

$$\boldsymbol{\theta} = (\alpha_{M2}, \ldots, \alpha_{Ma}) \text{ or } (\beta_{M2}, \ldots, \beta_{Mb}).$$

  In the general exploring of interaction effects,

$$\boldsymbol{\theta} = (\xi_{ij})_{i=2,\ldots a; j=2,\ldots,b}.$$

S2.2. General exploring: Scheffe's solution

- For an individual contrast $\boldsymbol{a}^\top \boldsymbol{\theta}$, we can consider the test statistic

$$T_{\boldsymbol{a}} = \frac{\boldsymbol{a}^\top \hat{\boldsymbol{\theta}}}{\sqrt{\boldsymbol{a}^\top \hat{\Sigma}_\theta \boldsymbol{a}}},$$

  where $\hat{\boldsymbol{\theta}}$ is the estimate of $\boldsymbol{\theta}$ and $\hat{\Sigma}_\theta$ is the estimated variance matrix of $\hat{\boldsymbol{\theta}}$.
- We need to find the critical value c_α such that

$$P(\cup_{all}\ \boldsymbol{a}\{|T_{\boldsymbol{a}}| \geq c_\alpha\}|H_0) \leq \alpha.$$

- To control the family-wise type I error rate at level α, the Scheffe's solution is to take

$$c_\alpha = \sqrt{m_1 F_{m_1,m_2}(\alpha)},$$

  where m_1 is the number of components of $\boldsymbol{\theta}$ and m_2 is the d.f. of $\hat{\sigma}^2$, $F_{m_1,m_2}(\alpha)$ is the upper α quantile of the F_{m_1,m_2} distribution. The critical value $c_\alpha = \sqrt{m_1 F_{m_1,m_2}(\alpha)}$ is referred to as the Scheffe's criterion.

§2.3. Derivation of Scheffe's criterion

- Note that

$$|T_{\boldsymbol{a}}| \geq c_\alpha \Leftrightarrow \frac{\boldsymbol{a}^\top \hat{\theta}\hat{\theta}^\top \boldsymbol{a}}{\boldsymbol{a}^\top \hat{\Sigma}_\theta \boldsymbol{a}} \geq c_\alpha^2.$$

$$\cup_{all\ \boldsymbol{a}}\{|T_{\boldsymbol{a}}| \geq c_\alpha\} = \{\max_{all\ \boldsymbol{a}} |T_{\boldsymbol{a}}| \geq c_\alpha\} = \left\{\max_{all\ \boldsymbol{a}} \frac{\boldsymbol{a}^\top \hat{\theta}\hat{\theta}^\top \boldsymbol{a}}{\boldsymbol{a}^\top \hat{\Sigma}_\theta \boldsymbol{a}} \geq c_\alpha^2\right\}.$$

- By the results of matrix theory (see Lecture notes 0),

$$\max_{all\ \boldsymbol{a}} \frac{\boldsymbol{a}^\top \hat{\theta}\hat{\theta}^\top \boldsymbol{a}}{\boldsymbol{a}^\top \hat{\Sigma}_\theta \boldsymbol{a}} = \lambda_{\max}(\hat{\theta}\hat{\theta}^\top \hat{\Sigma}_\theta^{-1}) = \hat{\theta}^\top \hat{\Sigma}_\theta^{-1}\hat{\theta} = \frac{\hat{\theta}^\top \Sigma_\theta^{-1}\hat{\theta}}{\hat{\sigma}^2/\sigma^2},$$

which, dividing by m_1, follows F_{m_1,m_2}. Thus,

$$c_\alpha^2 = m_1 F_{m_1,m_2}(\alpha), \text{ i.e., } c_\alpha = \sqrt{m_1 F_{m_1,m_2}(\alpha)}.$$

§2.4. Simultaneous p-values

- Recall that the p-value is the type I error rate when the observed test statistic value is taken as if it is the critical value.
- Let
$$H_0 : \boldsymbol{a}^\top \boldsymbol{\theta} = 0,$$
and $T_a = \frac{\boldsymbol{a}^\top \hat{\boldsymbol{\theta}}}{\sqrt{\boldsymbol{a}^\top \hat{\Sigma}_\theta \boldsymbol{a}}}$. The simultaneous p-value for testing H_0 is given by
$$P\left(\frac{\hat{\boldsymbol{\theta}}^\top \Sigma_\theta^{-1} \hat{\boldsymbol{\theta}}}{\hat{\sigma}^2/\sigma^2} \geq T_a^2\right) = P(F_{m_1,m_2} \geq T_a^2/m_1).$$

§2.5. Individual and simultaneous confidence intervals

- Let $[L_j,\ U_j]$ be an interval estimate of θ_j, $j = 1, \ldots, m$. An individual C.I. for θ_j with confidence coefficient $1 - \alpha$ only requires that

$$P(L_j \leq \theta_j \leq U_j) \geq 1 - \alpha,$$

  regardless whether the other intervals cover the other parameters.
- Simultaneous C.I. with confidence coefficient $1 - \alpha$ for all j require that

$$P\left(\cap_{j=1}^{m}\{L_j \leq \theta_j \leq U_j\}\right) \geq 1 - \alpha.$$

- Suppose we have 100 samples and for each sample construct 95% confidence intervals for two parameters: θ_1, θ_2.
  - If they are constructed as individual confidence intervals then there are roughly 95 samples producing intervals that cover θ_1, roughly another 95 samples producing intervals that cover θ_2. These two sets of samples are not necessarily the same.
  - If they are constructed as simultaneous intervals then there are roughly 95 common samples from which the constructed intervals cover θ_1 and θ_2 simultaneously.

§2.6. Scheffe's simultaneous confidence intervals

- With $c_\alpha = \sqrt{m_1 F_{m_1,m_2}(\alpha)}$, we have

$$\begin{aligned} & P\left(\cap_{all\boldsymbol{a}}\left\{\frac{|\boldsymbol{a}^\top\hat{\boldsymbol{\theta}} - \boldsymbol{a}^\top\boldsymbol{\theta}|}{\sqrt{\boldsymbol{a}^\top\hat{\Sigma}_\theta\boldsymbol{a}}} \le c_\alpha\right\}\right) \\ = \; & 1 - P\left(\cup_{all\boldsymbol{a}}\left\{\frac{|\boldsymbol{a}^\top\hat{\boldsymbol{\theta}} - \boldsymbol{a}^\top\boldsymbol{\theta}|}{\sqrt{\boldsymbol{a}^\top\hat{\Sigma}_\theta\boldsymbol{a}}} \ge c_\alpha\right\}\right) \\ = \; & 1 - \alpha. \end{aligned}$$

- *Simultaneous confidence intervals* for the contrasts $\boldsymbol{a}^\top\boldsymbol{\theta}$ are constructed as

$$\boldsymbol{a}^\top\hat{\boldsymbol{\theta}} \pm \sqrt{\boldsymbol{a}^\top\hat{\Sigma}_\theta\boldsymbol{a}}\sqrt{m_1 F_{m_1,m_2}(\alpha)}.$$

§2.7. Multiple comparison for main-effects

- We take the one-way ANOVA model for illustration. The regression model is given by
$$y_i = \beta_0 + \beta_2 u_{i2} + \cdots + \beta_a u_{ia} + \epsilon_i, \ i = 1, \ldots, n,$$
where the parameters have the meaning: $\beta_k = \mu_k - \mu_1$, $k \geq 2$.
- For any contrast, we have
$$\sum_{k=1}^{a} c_k \mu_k = \sum_{k=2}^{a} c_k \beta_k.$$
- The test statistic for testing $H_0 : c_k \beta_k = 0$ is given by
$$T_C = \frac{\sum_{k=2}^{a} c_k \hat{\beta}_k}{\sqrt{\tilde{\boldsymbol{c}}^\top \hat{\Sigma}_\beta \tilde{\boldsymbol{c}}}},$$
where $\tilde{\boldsymbol{c}} = (c_2, \ldots, c_a)^\top$, $\hat{\Sigma}_\beta$ is the estimated covariance matrix of $(\hat{\beta}_2, \ldots, \hat{\beta}_a)$.
- The Scheffe's criterion is given by $c_\alpha = \sqrt{(a-1)F_{a-1,n-a}(\alpha)}$.

§2.8. General exploring in action

- For a general exploring, there are no pre-determined contrasts to investigate, the task is to find the significant contrasts, if there is any.
- The general exploring is carried out in two steps:
  - (i) Conduct an overall significance test to see whether there is any effect of the treatments;
  - (ii) When the significance test is significant, find particular significant effects.
- It is impossible to investigate all possible contrasts. The contrasts under screening are to be identified from the summary results on estimated regression coefficients. Usually, the contrasts are those which are apparently significant.

§2.9. Meat storage (MS) example

- *Objective:* Study whether some form of controlled gas atmosphere would provide a more effective packaging environment for meat storage.
- *Response:* Psychrotrophic bacteria after 9 days.
- Packaging conditions: "Commercial plastic wrap"; "Vacuum package"; "1% CO, 40% O_2, 59% N"; "100% CO_2".

Packaging Condition	Psychorotrophic Bacteria Log(count/cm^2)		
Commercial plastic wrap	7.66	6.98	7.80
Vacuum packaged	5.26	5.44	5.80
1% CO, 40% O_2, 59% N	7.41	7.33	7.04
100% CO_2	3.51	2.91	3.66

§2.10. MS example: R codes and fitted result

```
y=c(7.66, 6.98, 7.80, 5.26, 5.44, 5.80,
    7.41, 7.33, 7.04, 3.51, 2.91, 3.66)
package = factor(kronecker(c(1:4),rep(1,3)))
fit = lm(y~package);summary(fit); anova(fit)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.4800     0.1965  38.064 2.49e-10
package2     -1.9800     0.2779  -7.125 9.95e-05
package3     -0.2200     0.2779  -0.792    0.451
package4     -4.1200     0.2779 -14.825 4.22e-07

Analysis of Variance Table
          Df Sum Sq Mean Sq F value    Pr(>F)
package    3 32.873 10.9576  94.584 1.376e-06
Residuals  8  0.927  0.1159
```

§2.11. MS example: General exploring

- The overall significance F-test has a p-value 1.376e-06. The effect of the packaging on storage is significant.
- The significance of the F-test does not tell what the specific effects are. To explore the possible significant contrasts, we can identify from the summary table the following apparently significant ones:

Contrasts in terms of μ_k's	Contrasts in terms of β_k's
$\mu_2 - \mu_1$	β_2
$\mu_4 - \mu_1$	β_4
$\mu_2 - \mu_3$	$\beta_2 - \beta_3$
$\mu_2 - \mu_4$	$\beta_2 - \beta_4$
$\mu_3 - \mu_4$	$\beta_3 - \beta_4$

§2.12. MS example: Multiple comparison results

- R codes for computation:

```
b = fit$coef[-1];V = vcov(fit)[-1,-1]
C = matrix(c(1,0,0, 0,0,1, 1,-1,0, 1,0,-1, 0,1,-1),
           ncol=3,byrow=T)
L=C%*%b; VC=C%*%V%*%t(C); sd=sqrt(diag(VC)); T=L/sd
p.value= pf(T^2/3,3,8,lower.tail=FALSE)
```

- Results:

Contrast	$\hat{C}$	sd($\hat{C}$)	T	p.value
β_2	-1.98	0.2779089	-7.124638	7.980780e-04
β_4	-4.12	0.2779089	-14.825004	3.691492e-06
$\beta_2 - \beta_3$	-1.76	0.2779089	-6.333012	1.752161e-03
$\beta_2 - \beta_4$	2.14	0.2779089	7.700366	4.675066e-04
$\beta_3 - \beta_4$	3.90	0.2779089	14.033378	5.624686e-06

- Conclusion: All the identified contrasts are significant. Package 4 is the best, followed by package 2. The difference between package 4 and 2 is significant. Package 1 is the worst. Package 3 and 1 do not have significant difference.

§3.1. Pairwise contrasts

- In certain practical problems, only the difference in certain pairs or all the pairs are of interest. Tukey's procedure is for the comparison of all pairs. Dunentt's procedure is for the comparison of all the treatments with a benchmark treatment.
- In terms of β_i's, the pairwise contrasts are of the forms:

$$\begin{aligned} \mu_i - \mu_1 &= \beta_i, \ i > 1; \\ \mu_i - \mu_j &= \beta_i - \beta_j, \ i, j > 1. \end{aligned}$$

- In Tukey's procedure, we seek c_α such that

$$P\left(\max_{k,j} |T_{ij}| \geq c_\alpha\right) \leq \alpha,$$

where

$$T_{ij} = \begin{cases} \frac{\hat{\beta}_i}{s.d(\hat{\beta}_i)}, & \text{if } j = 1; \\ \frac{(\hat{\beta}_i - \hat{\beta}_j)}{[\mathrm{Var}(\hat{\beta}_i) + \mathrm{Var}(\hat{\beta}_j) - 2\mathrm{Cov}(\hat{\beta}_i, \hat{\beta}_j)]^{1/2}}, & \text{otherwise.} \end{cases}$$

§3.2. Studentized range distribution

Let $\bar{Y}_i, i = 1, \cdots, a$ be the sample means of a samples with equal sizes, i.e., $n_i = n$. Let $\hat{\sigma}^2$ be the estimate of error variance σ^2.

- The studentized range statistic is defined as:

$$q_{a,n_{\mathrm{ERR}}} = \sqrt{n}(\max \bar{Y}_i - \min \bar{Y}_i)/\hat{\sigma}.$$

  Under the assumption that $\mu_1 = \cdots = \mu_a$, the distribution of $q_{a,n_{\mathrm{ERR}}}$ is referred to as the Studentized range distribution.

- Let $q_{a,n_{\mathrm{ERR}},\alpha}$ denote the upper α-quantile of the Studentized range distribution. The statistical table of $q_{a,n_{\mathrm{ERR}},\alpha}$ is given in the file `Studentized Range q Table.pdf`.
- Note that the first index a in $q_{a,n_{\mathrm{ERR}},\alpha}$ is the number of groups rather than the degrees of freedom.

§3.3. Studentized range statistic in terms of t-statistic

- The studentized range random variable can be expressed as

$$q_{a,n_{\mathrm{ERR}}} = \max_{ij} \sqrt{n}|\bar{Y}_i - \bar{Y}_j|/\hat{\sigma}. \tag{1}$$

- The t statistic T_{ij} for testing $H_0 : \mu_i - \mu_j = 0$, is the same as

$$T_{ij} = \sqrt{n/2}|\bar{Y}_i - \bar{Y}_j|/\hat{\sigma},$$

where n is replaced by $\tilde{n}_{ij} = \frac{2n_i n_j}{n_i + n_j}$ when $n_i \neq n_j$.

- The studentized range random variable is further expressed as

$$q_{a,n_{\mathrm{ERR}}} = \sqrt{2} \max_{ij} |T_{ij}|.$$

- Hence

$$\sqrt{n}|\bar{Y}_i - \bar{Y}_j|/\hat{\sigma} \geq q_{a,n_{\mathrm{ERR}},\alpha} \Leftrightarrow |T_{ij}| \geq q_{a,n_{\mathrm{ERR}},\alpha}/\sqrt{2}.$$

- The quantity $q_{a,n_{\mathrm{ERR}},\alpha}/\sqrt{2}$ is called the Tukey's criterion for t-test.

§3.4. Pairwise comparison with Tukey's criterion

- The pairwise contrast $\mu_i - \mu_j$ is claimed significant if

$$|T_{ij}| \geq q_{a,n_{\mathrm{ERR}},\alpha}/\sqrt{2}.$$

- The simultaneous comfidence intervals:
  - For $\mu_i - \mu_1$:

$$\hat{\beta}_i \pm sd(\hat{\beta}_i) q_{a,n_{\mathrm{ERR}},\alpha}/\sqrt{2}.$$

  - For $\mu_i - \mu_j, i, j, > 1$:

$$(\hat{\beta}_i - \hat{\beta}_j) \pm \left[\mathrm{Var}(\hat{\beta}_i) + \mathrm{Var}(\hat{\beta}_j) - 2\mathrm{Cov}(\hat{\beta}_i, \hat{\beta}_j)\right]^{1/2} q_{a,n_{\mathrm{ERR}},\alpha}/\sqrt{2}.$$

§3.5. MS example: pariwise comparison

- R codes

```
C = matrix(c(1,0,0,0,1,0,0,0,1,1,-1,0,1,0,-1,
             0,1,-1),ncol=3,byrow=T)
L = C%*%b; VC = C%*%V%*%t(C); sd = sqrt(diag(VC));T = L/sd
cbind(L, sd, T)
```

- The computed values:

```
  pair     C       s.C           T
2 vs 1 -1.98 0.2779089  -7.1246380 *
3 vs 1 -0.22 0.2779089  -0.7916264
4 vs 1 -4.12 0.2779089 -14.8250042 *
2 vs 3 -1.76 0.2779089  -6.3330115 *
2 vs 4  2.14 0.2779089   7.7003663 *
3 vs 4  3.90 0.2779089  14.0333778 *
```

$q_{4,8,0.05}/\sqrt{2} = 4.529/\sqrt{2} = 3.202487.$

- Conclusion: Except the difference between packages 3 and 1, all the other differences are significant.

§3.6. MS example: Tukey's confidence intervals

- R codes

```
cbind(L - sd*4.529/sqrt(2), L +sd*4.529/sqrt(2))
```

- Intervals

pair	lower limit	upper limit
2 vs 1	-2.869999	-1.0900006
3 vs 1	-1.109999	0.6699994
4 vs 1	-5.009999	-3.2300006
2 vs 3	-2.649999	-0.8700006
2 vs 4	1.250001	3.0299994
3 vs 4	3.010001	4.7899994

§3.7. Dunnett's criterion

- In certain practical problems, it is of interest to compare several new treatments with a standard treatment. Only the following contrasts are of concern:
$$\beta_j = \mu_j - \mu_1,\ j = 2, \ldots, a.$$
The Dunnett's procedure is for the comparison of these contrasts.
- Let $T_j = \hat{\beta}_j / sd(\hat{\beta}_j)$. The Dunnett's criterion, $d_{a,\nu,\alpha}$, is determined by
$$P(\max_{j=2,\ldots,a} |T_j| \geq d_{a,\nu,\alpha}) = \alpha.$$
where a is the number of groups including the control group, ν is the d.f. of the residuals.
- The $d_{a,\nu,\alpha}$ values, which assumes that $n_1 = \cdots = n_a$, are given in the file `Dunnetts-Table.pdf` provided in Canvas.
- For unequal group sizes, for contrast β_j, the criterion is adjusted by a factor $m_j = 1 + 0.07(1 - n_j/n_1)$, e.g., if the tabled criterion is $d_{a,\nu,\alpha}$, the adjusted criterion is $m_j d_{a,\nu,\alpha}$.

§3.8. Comparison procedure

- For each β_j, if $|T_j| > m_j d_{a,\nu,\alpha}$, β_j is claimed significant at level α.
- The 100(1-α)% simultaneous confidence intervals for the $a-1$ contrasts are constructed as

$$\hat{\beta}_j \pm sd(\hat{\beta}_j) m_j d_{a,\nu,\alpha}.$$

- In the MS example, the T_j statistics are given in the following:

```
pair      C        s.C            T
2 vs 1 -1.98 0.2779089  -7.1246380 *
3 vs 1 -0.22 0.2779089  -0.7916264
4 vs 1 -4.12 0.2779089 -14.8250042 *
```

$d_{4,8,0.05} = 2.880$. Thus β_2 and β_4 are significant at level $\alpha = 0.05$, i.e., the second and the fourth group are significantly different from the first group at level $\alpha = 0.05$.

§4.1. Bonferroni's method

Suppose there are k prespecified contrasts which are the only concern of the investigator. The overall type I error rate α for the k contrasts can be controlled by Bonferroni's method.

- Bonferroni's method control the overall type I error rate by adjusting the individual type I error rate. If the overall rate is α, say, for contrast j, the rate is controlled at $\alpha_j (< \alpha)$, subject to
$$\sum_{j=1}^{k} \alpha_j = \alpha.$$
- The allocation of the α_j's can be determined by consideration of the seriousness of the type I error associated with each of the contrasts. In general, α_j is taken equal for all j, i.e.,
$$\alpha_j = \alpha / k.$$

§4.2. The rationale of Bonferroni method

- Let T_j be the t-statistic for the jth contrast and c_{α_j} be the critical value such that $P(|T_j| \geq c_{\alpha_j}) \leq \alpha_j$. To control the overall type I error rate at a level α, the α_j's must be chosen such that
$$P(\cup_{j=1}^{k}\{|T_j| \geq c_{\alpha_j}\}) \leq \alpha.$$
- By the property of probability,
$$P(\cup_{j=1}^{k}\{|T_j| \geq c_{\alpha_j}\}) \leq \sum_{j=1}^{k} P(|T_j| \geq c_{\alpha_j}) \leq \sum_{k=1}^{j} \alpha_j.$$
Thus, if α_j's are chosen such that $\sum_{j=1}^{k} \alpha_j = \alpha$, the overall type I error rate is controlled at level α.
- Remark: The Bonferroni method is conservative. The actual type I error rate controlled by the method could be much smaller than the nominal level α, especially, when k is large.

§4.3. The critical value and simultaneous p-values

- If the overall type I error rate is to be controlled at level α, the critical value for each individual test is given by $t(\alpha/k/2)$.
- The p-value is the overall Type I error rate when the critical value is set at the observed value of the test statistic. For the jth test, if the observed value is T_j^0, its p-value is the overall type I error rate when the critical value is set at T_j^0. With this "critical value", the overall type I error rate is $kP(T_j \geq T_j^0)$, where the probability is computed under the distribution of T_j with the null hypothesis.
- The Bonferroni simultaneous p-values for k pre-determined contrasts are k times the individual p-values.

§4.4. Kenton Food Company example revisited

- In the Kenton Food Company example considered in Lecture Notes 3, the effects of cereal package on sales are investigated. More details on the four package designs are: (1) 3 colors with cartoons, (2) 3 color without cartoons, (3) 5 colors with cartoons, (4) 5 colors without cartoons.
- The primary concern of the company is whether or not there is differences between 3 colors and 5 colors, and between cartoons and no cartoons.
- Formulated in terms of contrasts, the primary concern is whether the following contrasts are nonzero:

$$\frac{\mu_1 + \mu_2}{2} - \frac{\mu_3 + \mu_4}{2}; \quad \frac{\mu_1 + \mu_3}{2} - \frac{\mu_2 + \mu_4}{2}.$$

- These are pre-specified contrasts, the Bonferroni's method is appropriate for the testing of these contrasts.

§4.5. KFC example: testing of pre-specified contrasts(1)

- Using the regression approach, we get the estimated regression coefficients and their variance matrix as follows:

$$\hat{\boldsymbol{\beta}} = (-1.2, 4.9, 12.6)^\top$$

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \begin{pmatrix} 4.218667 & 2.109333 & 2.109333 \\ 2.109333 & 4.746000 & 2.109333 \\ 2.109333 & 2.109333 & 4.218667 \end{pmatrix}.$$

- In terms of β_j's, the contrasts are:

$$\beta_2 - \beta_3 - \beta_4, \quad \beta_3 - \beta_2 - \beta_4;$$
$$\text{or} \quad \boldsymbol{c}_1^\top \boldsymbol{\beta}, \quad \boldsymbol{c}_2^\top \boldsymbol{\beta},$$

where $\boldsymbol{c}_1 = (1, -1, -1)^\top, \boldsymbol{c}_2 = c(-1, 1, -1)$

§4.6. KFC example: testing of pre-specified contrasts (2)

- The computation is done by the R codes:

```
c1=c(1,-1,-1); c2=c(-1,1,-1)
L1=t(c1)%*%b; s1=sqrt(t(c1)%*%V%*%c1); T1= L1/s1
L2=t(c2)%*%b; s2=sqrt(t(c2)%*%V%*%c2); T2= L2/s2
T=c(T1,T2)
2*2*pt(abs(T),15,lower.tail=FALSE)
```

- The test statistics: $T_1 = -6.245605$, $T_2 = -2.170932$.
- The Bonferroni p-values: `3.135027e-05`, `9.278879e-02`
- If the type I error rate is controlled at $\alpha = 0.05$, the first contrast is significant, but not the second one. It is concluded that there is a significant difference between 3 color design and 5 color design, but the difference between cartoon and no cartoon is not significant.

§4.7. Other application of Bonferroni's method

- The Bonferroni's method can also be used for many other situations:
  - (i) The individual tests can be of other types such as χ^2-tests, F-tests and so on;
  - (ii) The individual tests can be of different types;
  - (iii) The individual tests can themselves be families of tests.

  If there are k tests (or family of tests), the type I error rate for each test (or each family of tests) is adjusted to α/k, where α is the overall type I error rate.
- Suppose in a two-way ANOVA model we want to explore the effects of one factor at each level of the other factor. We can control the type I error rate at α/k within each level by Scheffe's method, if there are k levels of the other factor, in order to control the overall error rate at α.